FUTURE HEALTH | A GLOBAL INITIATIVE BY ABU DHABI | H2S

Future Health Digital Innovation Challenge

From signals to solutions.

Co-creator Microsoft

Co-creator inception42

Co-creator Mohamed bin Zayed University of Artificial Intelligence

Team Name: Silah · صلة
Team Size: 1
Problem Statement: Fertility · Challenge 03, Understanding and Reducing Inherited Disease Risk

Product page, demo and film

A private companion that helps Abu Dhabi families act on inherited disease risk before the next pregnancy, at birth and across the family, without ever deciding for them.

Team members*

A solo entry, built end to end by one person

S.No	Name	Country	Role	Organization	Age confirmation
1	Marc Donovici	France	Solo: product, engineering, research	Independent	Over 18: confirmed

A solo entry. Every part of Silah, from the risk arithmetic and the cryptography to the Arabic and English interface and the simulated health-system side, was designed, built and tested by one person.

Team Capability Summary*

Offline clinical AI, built and audited by the same person

I audit AI systems for a living and build offline clinical AI.

I lead AI for the Internal Audit function of a large European banking group, where I also audit information systems against GDPR and the EU AI Act. One mission covered the group's own AI architecture and its governance and security controls.

In clinical AI, FieldScreen AI, an offline tuberculosis screening tool for community health workers, placed 3rd of 850+ teams in Google's MedGemma Impact Challenge 2026.

For Silah I wrote the risk kernel, the verifier, the cryptography, the phone app and the simulated health-system side, and ran the model evaluation.

3rd of 850+

Google MedGemma Impact Challenge 2026, offline TB screening

Finalist, 1 of 6

European Patent Office CodeFest 2026, explainable ML

Track winner

EdgeAgent track, Global AI Hackathon with Qwen Cloud

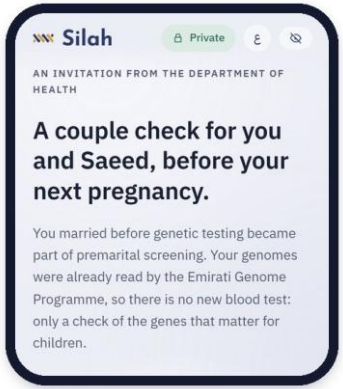
Architecture or workflow diagram*

One phone app, five services, only signed or blinded messages



Target users and intended outcome*

Three moments where knowing the risk still changes an outcome



Before the next pregnancy

Couples married before the mandate, invited to a couple check on genomes already held.

Outcome: an at-risk couple knows before the next pregnancy and reaches a counsellor.



At birth

Any at-risk couple expecting a baby, whatever they chose before.

Outcome: the known change is tested in the first days, ahead of the 21-day newborn genome report, so SMA or Krabbe is treated before signs.



In the family

Relatives of every carrier, invited without saying who sent the invitation.

Outcome: relatives come to testing early, with a reason, and nobody is exposed.

15,628
Emirati births in Abu Dhabi in 2024

3 in 4
or more of 2026 Emirati births are to couples the genetic test never reached (estimate)

900,000+
Emirati genomes already sequenced

565 / 11,748
couples found incompatible since 2022

How will success be measured?*

Measured on the DoH's KPI and on understanding, never on choices

Measure	How it is counted
Non-compatible couples referred to counselling	DoH KPI, target 90%; live in the pathway view
Days from result to counselling	Median, by language
Understanding: the four teach-back points	Asked by a researcher at the visit; Silah vs the current report, randomised by week
"Compatible means zero risk" misconception	Among compatible couples, both arms
Married couples who accept the couple check	Per invitation sent, by declared signal
At-risk pregnancies with a birth plan	Newborns tested in days 1–3, from maternity records
Relatives tested per invitation	Redeemed tokens; anonymous by design
Equity	Every measure by language and reading confidence; counts under 5 suppressed

Not a success measure

Marriages cancelled, or any single reproductive choice. Choices are reported, never ranked.

Already tracked in the prototype

Referral KPI, pathway funnel, teach-back shared, birth plans active, invitations redeemed, consent withdrawn.

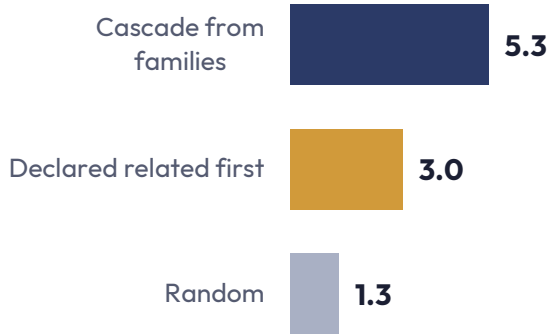
Evidence supporting the approach*

Exact arithmetic, privacy held end to end, outreach order tested

Tested in the prototype

- 175 unit tests against values worked out by hand, shared ancestry included: about 1 in 160 for Fatima's panel, where the certificate says "no risk".
- 30-check story through the real interface: neither partner's private finding reaches the other or the clinic.
- 0 WCAG 2.2 AA violations on 18 screens, both languages.
- Offline on an Android 13 emulator in airplane mode.
- Readable: English at US grade 6.4, against 14.2 for the DoH sheet couples sign today; Arabic 83.4 against 67.4 on OSMAN (higher is easier).

Which couples to invite first



At-risk couples found per 100 tests, after 200 tests. Simulated consanguineous population, 5 seeds, published Emirati carrier frequencies for 7 genes.

Safe answers, checked

52 questions	Grounded	Traps	Steers
Gemma E4B	22/22	28/30	0
Gemma E2B	21/22	30/30	0
Qwen 4B	22/22	30/30	1

22 grounded questions, 30 traps, Arabic and English. Steers: directive sentences written, all blocked. The phone shows the cited cards' fixed words, not the model's: then 30/30 traps answered.

- Early SMA treatment: 22 of 25 babies treated before signs later walked (De Vivo 2019).
- Official compatible-couple wording: "No risk for either the other partner or children to be carrier or diseased" (DoH Standard V3, 2024).

Data sources and legal right to use them*

No personal data: public standards, published figures, open licences

Source	Owner	Used for	Right to use	Personal data
Premarital Screening Standard V3 (2024)	DoH Abu Dhabi	Panel (570 printed names = 563 genes), report wording, 90% KPI	Public standard, cited	No
HGNC gene symbols	HUGO Gene Nomenclature Committee	Mapping the panel to current symbols	CC0	No
Jain et al. 2025; Rabea et al. 2024	Authors / publishers	Emirati carrier frequencies (7 genes)	Published facts, cited	Aggregate
NURTURE, SPR1NT and other trials	Authors / journals	Birth-plan statements	Published facts, cited	Aggregate
Programme figures (news reports)	Gulf News, Aletihad, Gulf Today	565 of 11,748; 900,000 genomes; births	Published facts, cited	No
Gemma 4 E2B / E4B	Google	Optional: picks checked answers on the device	Apache-2.0	No
Neural voices (ar-AE, en-GB)	Microsoft	Pre-recorded narration	Prototype use	No
Family, genomes, clinic records	Written for this prototype	The whole demonstration	Own work	Fictional

In a deployment, genomic data stays with the lab and on the person's phone, and services are hosted in the UAE (Federal Law No. 2 of 2019). Couple results from EGP genomes need fresh consent, which Silah collects.

Risks and safeguards*

Every risk has a safeguard in code, and a person is one tap away

Risk	What could go wrong	Safeguard in Silah
Privacy	A partner or relative learns a finding	Each partner sees only shared genes; sharing is by choice, signed, revocable; invitations name no sender
Security	A forged result or tampered consent	Lab results Ed25519-signed and checked on the phone; receipts hash-chained; vault encrypted
Bias	Outreach that profiles families	Ranking uses declared signals only, never genetics; every measure by language; counts under 5 suppressed
Explainability	Numbers nobody can check	Every number is an exact fraction with its arithmetic and source; assumptions labelled
Misuse	Steering a decision, or a wrong number	People read fixed, checked sentences only; the model just picks them; options at equal weight, never alone
Human oversight	Distress, religion, termination	Hard routes to a person before any model; “seen” only once the clinic confirms; quick hide
Failure modes	Model fails or omits a risk	Common risk questions answered from fact cards; model optional; works offline

Hardware requirements

No hardware to buy: Silah runs on the phones people already have

Phone

Any recent browser. Installs as an app, works offline, about 7 MB stored for offline use. Arabic and English, larger text.

Optional model on the phone

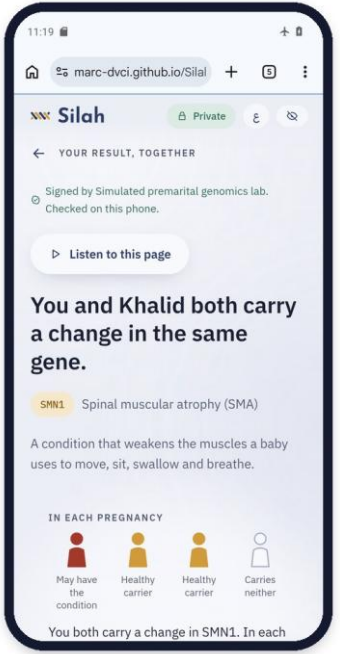
Gemma 4 E2B in the browser on WebGPU, a one-time download of about 3 GB. Without it, common questions still work.

Counsellor computer

Gemma 4 E4B through llama.cpp. Measured on an RTX 4070: median 1.8 s per answer.

Health-system services

Standard servers hosted in the UAE. No sensor, wearable or device touches the body, so there is no device safety risk.



Android 13, airplane mode: the live app reopened offline

Path to pilot and scale*

A pilot at one clinic, then a priced path to every married couple

Stakeholder	Role
DoH Abu Dhabi	Owns the programme; approves the check, outreach and result format
M42 · EGP	Holds the genomes, runs the couple check, signs results
SEHA · Corniche Hospital	Counselling clinic, maternity, IVF + PGT
Premarital centres	Issue the signed result with the certificate

Regulation: DoH Responsible AI Standard (2025), mapped clause by clause; treated as high risk, so a conformity assessment comes first. UAE health data law, data in the UAE.

Who pays: the programme owner, DoH or M42. Families never pay. No new sequencing, and no server cost per question: the model runs on the device. Timeline is an assumption.

Months 0–3	DoH approval of result format, consent and birth-plan text; Responsible AI conformity assessment; UAE hosting
Months 3–9	One centre, Corniche counselling and maternity; Silah vs current report, randomised by week
Months 9–18	Invite couples married before the mandate; birth plans across SEHA maternity units

Why DoH would pay

Measured: 9 genetic counsellors and 8 clinical geneticists for nearly 11 million people.

Assumed: 20,000 married couples re-checked a year find about 960 at risk, 8 times today's flow.

Priced: 20,000 re-checks plus about 2,600 premarital couples, at an assumed AED 100 each: about AED 2.26 million a year. Silah does the explaining and triage so those 17 people can counsel.

Pre-existing IP and third-party materials*

Written for this challenge; every third-party part is open-licensed

Silah's own work

Risk kernel, verifier, consent receipts, invitations, phone app, clinic side, fact cards, Arabic and English text, evaluation set and tests: written by Marc Donovici for this challenge, October–November 2026.

Public at github.com/Marc-Dvci/Silah under the Apache License 2.0.

Restrictions

No proprietary data. Narration voices need a commercial licence for production.

Component	Licence
Preact, Vite, Workbox (vite-plugin-pwa)	MIT
@noble/curves, @noble/hashes	MIT
@cloudflare/blindrsa-ts (RFC 9474)	Apache-2.0
Transformers.js; llama.cpp	Apache-2.0; MIT
Gemma 4 E2B / E4B weights	Apache-2.0
qrcode	MIT
IBM Plex Sans, Sans Arabic, Mono; Reem Kufi	SIL Open Font Licence
HGNC gene symbols	CC0
Microsoft neural voices (narration)	Prototype use

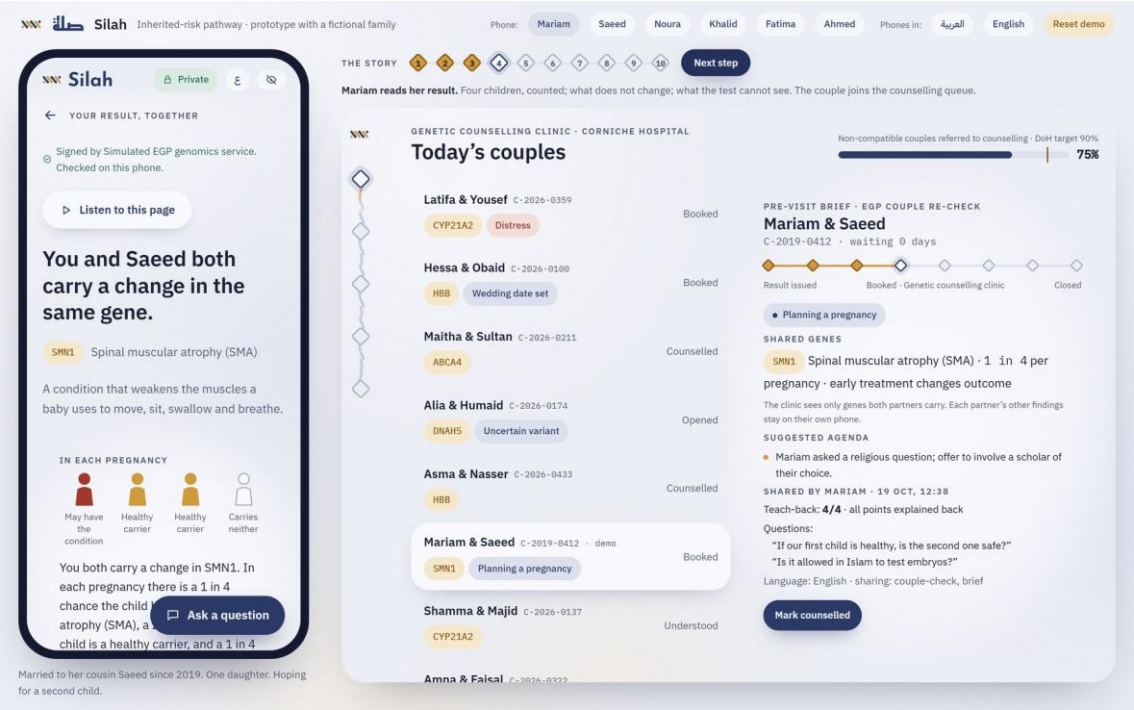
Conflict of interest declaration*

None. No member of the team has a conflict of interest to declare.

No	Employed by, or contracted to, the Department of Health – Abu Dhabi
No	Employed by, or contracted to, Future Health or a challenge collaborator (Microsoft, Inception42, MBZUAI, Hack2skill)
No	Family or close personal relationship with a judge, organiser or collaborator staff member
No	Financial interest in an organisation involved in running or judging the challenge
No	Access to non-public information about the challenge or its evaluation

Additional Slide

Try it: phone and health system side by side, in one browser



- Ten steps with a Next button; each step moves both screens and finishes itself.
- One switch puts every phone in Arabic or English.
- Fictional family: Mariam and Saeed (married 2019), Noura and Khalid (expecting), Fatima and Ahmed (engaged, "compatible").
- Runs offline after the first visit; no account, no server, no API key.
- marc-dvci.github.io/Silah

Scan for the product page, the live demo, the film and the documents.

Additional Slide

Assumptions, dependencies and limitations, each shown where it applies

Assumptions

- Carrier frequency of 1 in 1,000 for genes without an Emirati figure: the most cautious value the published added chance for cousins allows. 7 genes use published figures.
- Detection comes from the lab's signed result; the simulated lab states 90% (99% for the exact family change), and without it Silah shows ranges. From 80% to 99%, Fatima's 1 in 160 moves from 1 in 82 to 1 in 1,000.
- Re-check volume (20,000 couples a year), price (AED 100 per couple) and the pilot timeline.

Dependencies

- A DoH decision, with fresh consent from both partners, to return couple results from genomes EGP already holds. Silah provides the consent flow.
- A lab-signed result issued with the existing digital certificate; the integration with DoH, M42 and Malaffi is specified in the repository.
- Gemma 4 (Apache-2.0) for optional answers; common questions need no model.

Limitations of the prototype

- The family, their genomes and every clinic record are fictional; the lab, clinic, hospital and invitation service run in the browser.
- Understanding is measured on the text, in both languages; the pilot measures it in couples, randomised by week.
- Chances are model-derived and leave out family history. The model's own wording (52 answers per model, transcripts published) is never shown; people read fixed sentences.

Full list with sources: docs/ASSUMPTIONS_AND_LIMITATIONS.md at **github.com/Marc-Dvci/Silah**

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Thank you

Co-creator

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Mohamed bin Zayed University of Artificial Intelligence